

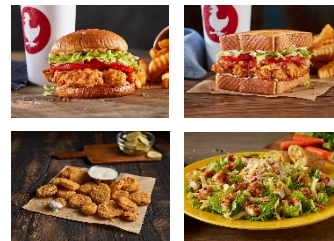
What are the Rules at Zaxby's?



Zaxby's is a chain of fast casual restaurants offering chicken wings, chicken fingers, sandwiches, and salads. Your family decides to order dinner using a food delivery app. The app lets you customize your order, track your delivery, and rate your experience.

1. You order the following items through the app for your family:

- Chicken Finger Sandwich with fries and a Diet Coke
- Zaxby's Club Sandwich with fries and a sweet tea
- The Zensation Salad with a Diet Coke
- Fried Pickles



List all **unique items** in this order.

set notation { Chicken finger sandwich, fries, Diet Coke, Zaxby's club sandwich, sweet tea, zensation salad, fried pickles }

2. After your order is delivered, the app asks you to rate your experience. You can give the order anywhere from 1 to 5 stars. List all possible ratings you could submit.

{ 1, 2, 3, 4, 5 }

3. The app guarantees your order will arrive in less than 20 minutes.

- a. Can your delivery take exactly 20 minutes?

NO

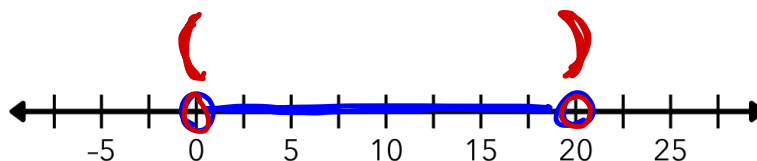
- b. Can your delivery take 7.238 minutes?

Yes

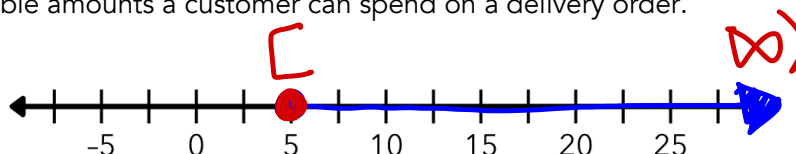
- c. List all possible delivery times or explain why this is not possible.

This is not possible because there are infinitely many options.

4. Use the number line below to show the possible delivery times for an order placed through the app.



5. The app requires a minimum order of \$5 for delivery. Use the number line below to show the possible amounts a customer can spend on a delivery order.



Lesson 0.7 – Set Notation

QuickNotes	{set}	[Interval]
Algebraic use inequalities to state possible values. Ex). $5 \leq x \leq 10$ $x \geq 15$ $x < -5$	List all unique items or write a rule. $\{x \in \mathbb{R} \mid x > 3\}$ x is an element of the reals such that	(lower bound, upper bound] • parentheses $\rightarrow$ not included • bracket $\rightarrow$ included • combine more than 1 interval with $\cup$ symbol • always use parentheses with ∞ .

Check Your Understanding

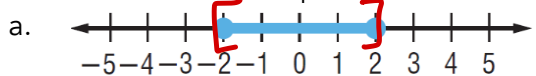
1. Five friends have the following birthdays: January 18, April 19, April 29, May 11, and May 21. List the **months** of the birthdays in set notation.

$\{\text{January, April, May}\}$

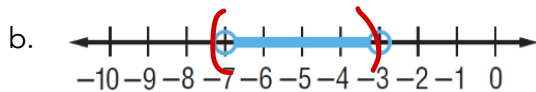
2. In set notation, list all multiples of three between 0 and 21, inclusive.

$\{0, 3, 6, 9, 12, 15, 18, 21\}$

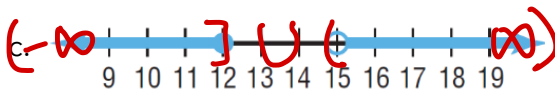
3. Use interval notation to represent the following quantities:



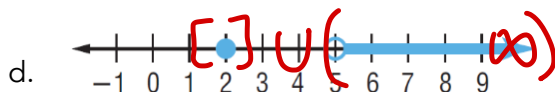
$[-2, 2]$



$(-7, -3)$



$(-\infty, 12] \cup (15, \infty)$



$\{2\} \cup (5, \infty)$

4. In order to receive bonus pay for summer work, Kelly has to work at least 60 hours in her online course. Due to other commitments, she is unable to work more than 120 hours. Use set notation **and** interval notation to represent the number of hours she can work in the online course.

$\{x \in \mathbb{R} \mid 60 \leq x \leq 120\}$ $[60, 120]$